

CUSTOMER SHOPPING INSIGHTS DASHBOARD

End-to-End Data Analytics Project Report

Python · SQL Server · Power BI

Prepared by: Bal Mahendran Sekar

Ajman, United Arab Emirates

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1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in the Review Rating column

Key Features

- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

3. Exploratory Data Analysis Using Python

Data preparation and cleaning were carried out in Python using the pandas library, following the steps below.

3.1 Import Libraries

```
import pandas as pd
```

3.2 Data Loading

```
DF = pd.read_csv('../Raw Data/customer_shopping_behavior.csv')
```

3.3 Initial Exploration

The dataset structure and summary statistics were reviewed using `DF.info()` and `DF.describe()`.

```
DF.info()  
DF.describe(include='all')
```

#	Column	Non-Null Count	Dtype
0	customer_id	3900	int64
1	age	3900	int64
2	gender	3900	str
3	item_purchased	3900	str
4	category	3900	str
5	purchase_amount (USD)	3900	int64

#	Column	Non-Null Count	Dtype
6	location	3900	str
7	size	3900	str
8	color	3900	str
9	season	3900	str
10	review_rating	3863	float64
11	subscription_status	3900	str
12	shipping_type	3900	str
13	discount_applied	3900	str
14	promo_code_used	3900	str
15	previous_purchases	3900	int64
16	payment_method	3900	str
17	frequency_of_purchases	3900	str

3.4 Column Standardization

Column names were renamed to snake_case for readability and consistency.

```
DF.columns = DF.columns.str.replace(' ', '_').str.lower()
DF.columns = DF.columns.str.replace('purchase_amount_(usd)',
'purchase_amount').str.lower()
```

3.5 Missing Data Handling

Null values were identified and the 37 missing entries in review_rating were imputed using the median rating of each product category, resulting in zero missing values across all columns.

```
DF.isnull().sum()
DF['review_rating'] = DF.groupby('category')['review_rating'].transform(lambda x:
x.fillna(x.median()))
DF.isnull().sum()
```

3.6 Feature Engineering

- Created an age_group column by binning customer ages.
- Created a purchase_cycle_days column derived from purchase frequency.

```
bins = [0, 19, 30, 60, 100]
labels = ['Teenager', 'Adult', 'Middle-Aged', 'Senior']
DF['age_group'] = pd.cut(DF['age'], bins=bins, labels=labels)
```

Age Group	Age Range
Teenager	18 – 19
Adult	20 – 30
Middle-Aged	31 – 60
Senior	61 – 70

```
freq_days_map = {
    'Weekly': 7, 'Fortnightly': 14, 'Bi-Weekly': 14, 'Monthly': 30,
    'Quarterly': 90, 'Every 3 Months': 90, 'Annually': 365
}
DF['purchase_cycle_days'] = DF['frequency_of_purchases'].map(freq_days_map)
```

Frequency of Purchase	Cycle (Days)
Weekly	7
Fortnightly	14
Bi-Weekly	14
Monthly	30
Quarterly	90
Every 3 Months	90
Annually	365

3.7 Data Integration

The cleaned DataFrame was loaded into an MSSQL Server database to support structured SQL analysis.

```
import pyodbc
from sqlalchemy import create_engine

server = 'AADHAN'
database = 'PowerBIDemo'
engine = create_engine(
    f'mssql+pyodbc://@{server}/{database}?
    driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'
)
DF.to_sql('customer_shopping_behavior', engine, if_exists='replace', index=False)
```

4. Data Analysis Using SQL (Business Transactions)

Structured analysis was performed to answer ten key business questions using SQL queries against the customer_shopping_behavior table.

4.1 Revenue by Gender

Business question: What is the total revenue generated by male vs. female customers?

```
SELECT gender AS Gender,
       CAST(SUM(CAST(purchase_amount AS DECIMAL(10,2))) AS DECIMAL(10,2)) AS
Revenue
FROM customer_shopping_behavior
GROUP BY gender;
```

Gender	Revenue
Male	157,890.00
Female	75,191.00

4.2 High-Spending Discount Users

Business question: Which customers used a discount but still spent more than the average purchase amount?

```
SELECT customer_id,
       CAST(purchase_amount AS DECIMAL(10,2)) AS purchase_amount
FROM customer_shopping_behavior
WHERE discount_applied = 'Yes'
   AND purchase_amount >= (SELECT AVG(purchase_amount) FROM
customer_shopping_behavior);
```

Sample of the result set (16 qualifying customers):

Customer ID	Purchase Amount
2	64.00
3	73.00
4	90.00
7	85.00
9	97.00
12	68.00
13	72.00
16	81.00
20	90.00
22	62.00
24	88.00
29	94.00
32	79.00
33	67.00

Customer ID	Purchase Amount
35	91.00
37	69.00

4.3 Top 5 Products by Rating

Business question: Which are the top 5 products with the highest average review rating?

```
SELECT TOP 5 item_purchased,
      ROUND(AVG(review_rating), 2) AS Review_Rating
FROM customer_shopping_behavior
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC;
```

Item Purchased	Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4.4 Shipping Type Comparison

Business question: How do average purchase amounts compare between Standard and Express shipping?

```
SELECT shipping_type,
      CAST(AVG(CAST(purchase_amount AS DECIMAL(10,2))) AS DECIMAL(10,2)) AS
purchase_amount
FROM customer_shopping_behavior
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Shipping Type	Avg. Purchase Amount
Standard	58.46
Express	60.48

4.5 Subscribers vs. Non-Subscribers

Business question: How does average spend and total revenue compare across subscription status?

```
SELECT subscription_status,
      COUNT(customer_id) AS Total_Customer,
      CAST(AVG(CAST(purchase_amount AS DECIMAL(10,2))) AS DECIMAL(10,2)) AS
Average_Spend,
      CAST(SUM(CAST(purchase_amount AS DECIMAL(10,2))) AS DECIMAL(10,2)) AS
Revenue
FROM customer_shopping_behavior
GROUP BY subscription_status;
```

Subscription Status	Total Customers	Avg. Spend	Revenue
Yes	1053	59.49	62,645.00
No	2847	59.87	170,436.00

4.6 Discount-Dependent Products

Business question: Which 5 products have the highest percentage of purchases made with a discount applied?

```
SELECT TOP 5 item_purchased,
      CAST(100 * SUM(CAST(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0 END AS
DECIMAL(10,2)))
      / COUNT(*) AS DECIMAL(10,2)) AS Discount_Rate
FROM customer_shopping_behavior
GROUP BY item_purchased
ORDER BY Discount_Rate DESC;
```

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4.7 Customer Segmentation

Business question: How can customers be segmented into New, Returning, and Loyal groups based on previous purchases?

```
WITH Customer_Segment AS (
  SELECT customer_id, previous_purchases,
        CASE WHEN previous_purchases <= 1 THEN 'New'
              WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
              ELSE 'Loyal' END AS Segment
  FROM customer_shopping_behavior)
SELECT Segment, COUNT(*) AS Total_Customers
FROM Customer_Segment
GROUP BY Segment;
```

Segment	Total Customers
Returning	701
Loyal	3116
New	83

4.8 Top 3 Products per Category

Business question: What are the top 3 most purchased products within each category?

```
WITH ItemCounts AS (
    SELECT category, item_purchased, COUNT(customer_id) AS Purchase_Count,
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)
                               DESC) AS RowNum
    FROM customer_shopping_behavior
    GROUP BY category, item_purchased
)
SELECT category, item_purchased, Purchase_Count
FROM ItemCounts
WHERE RowNum <= 3
ORDER BY category, RowNum;
```

Category	Item Purchased	Purchase Count
Accessories	Jewelry	171
Accessories	Belt	161
Accessories	Sunglasses	161
Clothing	Blouse	171
Clothing	Pants	171
Clothing	Shirt	169
Footwear	Sandals	160
Footwear	Shoes	150
Footwear	Sneakers	145
Outerwear	Jacket	163
Outerwear	Coat	161

4.9 Repeat Buyers & Subscriptions

Business question: Are customers with more than 5 previous purchases more likely to subscribe?

```
SELECT CASE WHEN subscription_status = 'Yes' THEN 'Subscription Customer'
           ELSE 'Non-Subscription Customer' END AS subscription_status,
       COUNT(customer_id) AS RepeatCustomer
FROM customer_shopping_behavior
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Subscription Status	Repeat Customers
Subscription Customer	958
Non-Subscription Customer	2518

4.10 Revenue Contribution by Age Group

Business question: What is the revenue contribution of each age group?

```
SELECT age_group,  
       SUM(purchase_amount) AS Revenue  
FROM customer_shopping_behavior  
GROUP BY age_group  
ORDER BY Revenue DESC;
```

Age Group	Revenue
Middle-Aged	132,638
Adult	48,199
Senior	43,164
Teenager	9,080

5. Dashboard in Power BI

An interactive Power BI dashboard was built to consolidate the analysis into a single visual summary, with slicers for subscription status, gender, category, and shipping type.



Figure 1: Customer Shopping Insights Dashboard (Power BI)

6. Business Recommendations

Boost Subscriptions

- Promote exclusive benefits for subscribers.

Customer Loyalty Programs

- Reward repeat buyers to move them into the “Loyal” segment.

Review Discount Policy

- Balance sales boosts with margin control.

Product Positioning

- Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

- Focus efforts on high-revenue age groups and express-shipping users.